

1. Product Overview

Product name: Jibble

Type: Mobile-first social media app for closed communities (primarily colleges/universities)

Platforms (v1): Android (Flutter), later iOS

Elevator pitch:

Jibble is a focused, non-addictive social media app for colleges and other closed communities where members can:

- Connect with real classmates and colleagues
- Post both openly and anonymously
- Join events and groups
- Match based on skills and interests

All within a safe, private, organization-centric environment rather than an open, global network.

Problem statement:

Existing social apps (Instagram, etc.) are global, addictive, and not optimized for closed communities such as colleges, schools, or companies. Students cannot easily:

- Interact only within their college community
- Share sensitive content anonymously but safely
- Discover events, groups, and opportunities centered around their campus
- Match with peers based on skills for projects, events, or hiring

High-level goals:

- Build a dedicated social layer for closed communities (starting with colleges).

- Reduce “doom-scrolling” by offering category-based, filterable feeds and community-focused content.
- Help creators reach a more relevant audience by category, community, and skills.
- Enable anonymous but controlled sharing within a trusted community.

2. Target Users and Use Cases

Primary audience:

- College and university students (UG/PG).
- Secondary: Schools, companies, organizations, any closed group with limited members (e.g., training institutes, bootcamps, hostels).

Key personas:

1. Student (Member) – “Aarav”

- Wants a place to see campus events, join groups, post memes, ask doubts, and vent anonymously.
- Needs safe, non-toxic, non-addictive environment.

2. Society/Class Representative / Community Admin – “Neha”

- Manages clubs, events, and groups.
- Needs tools to announce events, create groups, and keep discussions in one place.

3. Skill-based Creator / Mentor – “Rohan”

- Posts educational or skill-based content (coding, design, etc.).
- Wants a relevant audience, discovery based on category/skills, and maybe hiring/connect opportunities later.

Core use cases:

- Student joins Jibble, selects college, sets profile, and starts browsing a focused home feed.
- Student posts content (public or anonymous) for their college and followers.
- Student joins “Circle” to see college-specific anonymous posts, groups, events, and skill matching.
- Members chat 1:1 or in groups, coordinate events, and network skill-wise.
- Admins promote events, manage groups, and moderate content.

3. Scope and Non-Scope (v1)

In-scope (v1):

- Single college / organization per account (multi-org later).
- Auth, onboarding, college selection, profile creation.
- Feed with posts (text, images, possibly short video later).
- Category-based filtering (comedy, horror, academic, etc.).
- Likes, comments, shares, following.
- Anonymous posts within the college community.
- Circle (college-only): anonymous stream, events, groups, skill matching.
- Real-time chat (1:1 and basic groups).
- Stories (simple, images or short clips).
- Basic notifications.
- Basic moderation/reporting.

Out-of-scope (initially):

- Web app.
- Advanced creator monetization payouts.
- Complex recommendation engine (full ML).

- Cross-college discovery.
- Public, internet-wide content.

4. Core Features and Requirements

4.1 User Authentication

Description: Secure login and signup, supporting email/phone (and optionally college email).

Requirements:

- Signup with email/phone, password, and OTP verification.
- Optional verification via college domain email for “verified student” badge (future iteration).
- Login with credential + secure session tokens (JWT).
- Forgot password flow (email/OTP reset).
- Logout on all devices option.
- Enforcement of strong password policy and rate limiting on login attempts.

4.2 Onboarding and College/Organization Selection

Description: Map each user to one primary community (college/organization).

Requirements:

- On first login: collect name, username, college/university (or organization), year/department (optional), DOB.
- Searchable list of colleges/organizations; if not found, “Request to add” flow (added by Jibble team, not users).

- Users cannot create communities; only Jibble admins can create/approve communities.
- Option to select interests and categories (comedy, academics, sports, etc.) to tune default feed.

4.3 Profile Creation and Management

Description: Public-facing identity plus some private fields.

Requirements:

- Fields: name, username (unique), profile photo, bio, college, year/branch, skills/interests, optional social links.
- Show counts: followers, following, posts.
- Edit profile with validation and update history.
- Privacy options:
 - Public profile within community vs restricted (Followers-only).
 - Show/hide certain details (year, branch, etc.).
- View others' profiles (tabbed: Posts, Stories, Circle involvement, etc.).

4.4 Post Creation

Description: Create posts to share text/media to feed and/or Circle.

Requirements:

- Create post with: text, one or multiple images, optional category (comedy, academic, etc.), visibility (public-in-college, followers only, or Circle).
- Toggle for anonymous posting within college/Circle (identity hidden to other users but logged internally).
- Media upload to AWS S3; compression and upload progress.

- Ability to tag category and possibly location (college-specific).
- Draft saving (optional v1).
- Basic content validation (length limits, media size, content type).

4.5 Feed and Discovery

Description: Main consumption surface for content.

Requirements:

- Home feed:
 - Posts from followed users.
 - Relevant posts from same college community.
 - Suggested posts from creators based on interests/categories.
- Category filters:
 - Option to switch feed view by category (comedy, horror, academics, events, etc.).
 - Users can select multiple categories to focus the feed.
- “Not addictive” principles (v1 ideas):
 - No endless scroll; optional content “batch” pagination with “Show more” button.
 - Daily time insights; optional “time limit reminders”.
- Highlight posts from college-specific Circle items (events, top anonymous posts) in a separate section or tab.

4.6 Reactions: Likes, Comments, Shares, Follow

Description: Basic social interactions.

Requirements:

- Like/unlike posts (count visible).
- Comment on posts (threaded comments optional later).
- Share post internally (to chats, groups) and copy-link (if allowed).
- Follow/unfollow users.
- Block/mute users.
- For anonymous posts:
 - Allow like/comment but indicate “Anonymous” as author; user identity hidden from others.

4.7 Stories

Description: Ephemeral content similar to Instagram Stories but scoped to community.

Requirements:

- Post story (image/video up to N seconds).
- Story viewer UI with progress bar and swipe navigation.
- Stories visible for 24 hours.
- Option to post story to: followers only or entire college.
- Report story and mute story from a user.

4.8 Circle – College Community Hub

Description: A dedicated page/tab for each college community, containing four main areas: Anonymous, Events, Groups, Matching.

Circle root screen:

- Tabs or sections:
- Anonymous
- Events
- Groups
- Matching

4.8.1 Circle – Anonymous

Requirements:

- Anonymous post feed scoped to the user's college.
- Post anonymously with text and optional media.
- Reactions: likes, comments, report.
- Strong moderation: quick ability to hide content and penalize users internally.

4.8.2 Circle – Events

Requirements:

- List of upcoming and past events in the college.
- Event card: title, date/time, location, organizer, description, cover image, RSVP count.
- Create event:
 - Only approved roles (e.g., Jibble admin or college ambassador/club heads).
 - Fields for registration link and contact details.
- RSVP/Interested button and attendee list (counts visible, names optional).
- Send reminders via notifications (before event).

4.8.3 Circle – Groups

Requirements:

- Directory of groups (clubs, classes, project teams) for the college.
- Group types: public (anyone in college can join) or request-only.
- Group page: description, admins, members count, posts feed and chat link.
- Group chat or pinned chat link.
- Group creation:
 - Initially restricted to Jibble-approved roles (ambassadors/admins).
 - Users can request new group; Jibble team or campus ambassador approves.

4.8.4 Circle – Matching (Skills)

Requirements:

- Skill profile: each user can list skills and interests (e.g., Flutter, design, public speaking).
- Matching screen:
 - Swipe-style or list-style UI to discover people with complementary/interested skills.
 - Filters: skill, year, role (looking for project teammates, looking for internships, etc.).
- “Make card” screen:
 - User creates a short profile card with: skills, what they’re looking for (project team, startup partner, study group, etc.), short bio.
- Ability to send connection request or direct message from matching.

4.9 Real-Time Chat

Description: Messaging for 1:1 and groups.

Requirements:

- Chat list with pinned chats, unread indicators.
- 1:1 chat: text, emojis, images (and later voice notes).
- Group chat: created via groups in Circle or manually between multiple users.
- Real-time via WebSocket (Socket.io) or similar.
- Message status: sent, delivered, seen.
- Basic moderation:
 - Block users; no messages from blocked users.
 - Report chat message.
- Messages stored securely with retention policy (e.g., optional archive after 1 year).

4.10 Notifications

Description: Keep users updated without spamming.

Requirements:

- Types: new followers, likes, comments, mentions, event reminders, group invites, chat messages, skill match requests.
- In-app notification center with read/unread state.
- Push notifications with per-category toggle in settings.
- Do-not-disturb window option.

4.11 Search

Description: Discover users, groups, events, and posts.

Requirements:

- Search bar for: users, groups, events, skills, and categories.
- Filter by type and by community (within own college by default).
- Search results page with clear sections.

4.12 Admin & Moderation Tools (MVP)

Requirements:

- Role-based admin (global Jibble admin, college ambassador, group admin).
- Admin capabilities:
 - Delete or hide posts and comments.
 - Temporarily or permanently suspend accounts.
 - Manage event and group creation approvals.
- Abuse reporting dashboard (maybe basic API + admin web panel later).

4.13 Monetization (Phase-wise)

Overall principle: Monetization must not break trust or create spammy experience.

Features:

1. Ads (Later phase):

- Non-intrusive, category-relevant ads.
- Only in specific placements (between feed batches, event section, etc.).

2. Premium features:

- Verified creator profile.

- Advanced analytics for creators (reach, engagement by category).
- Extra visibility/boost in matching or events.

3. Event promotions:

- Paid promotion for events to appear higher in event lists or home suggestions within that college.

4. Skill-based hiring connections:

- For companies or senior students to pay for access to curated skill-based profiles (opt-in only from users).
- Separate access rules to protect student privacy.

5. App Flow and Navigation

High-Level App Flow:

1. Install app.
2. Splash screen with Jibble branding.
3. Authentication screen (login/signup).
4. Signup → OTP / verification → onboarding.
5. Onboarding steps:
 - Choose college/organization.
 - Enter personal details & username.
 - Select interests and skills.
 - Basic tutorial (optional).
6. Land on Home page (feed).
7. From bottom navigation or tabs, user can go to:
 - Home (feed)

- Circle (community)
- Post creation (central FAB/button)
- Chat
- Profile

8. From Circle, access Anonymous, Events, Groups, Matching.

9. From Home/Profile/Circle, user can tap into: post detail, story viewer, profiles, event details, group pages.

10. Settings accessible from Profile or top-right menu.

Navigation Structure:

- Bottom navigation (recommended):

- Home
- Circle
- Create Post (center FAB)
- Chat
- Profile

- Global actions:

- Search icon at top of Home and Circle.
- Notification bell at top-right.

6. Screen-by-Screen Requirements (UI/UX)

6.1 Splash Screen

- Jibble logo and tagline.

- Smooth short animation (max 2–3 seconds).
- Auto-navigate to Auth screen or Home if already logged in.

6.2 Authentication Screen

- Tabs or toggle: Login / Sign Up.
- Fields: email/phone, password; “Forgot password”; “Send OTP” if using phone.
- “Continue” button with clear disabled/enabled states.
- Links to Terms and Privacy Policy.

6.3 Onboarding Screens

- Step 1: Basic info (name, username, profile pic; validation for unique username).
- Step 2: College/organization selection (searchable dropdown).
- Step 3: Interests and categories selection (chips/tags).
- Step 4: Skills selection and optional short bio.
- Progress indicator for steps.

6.4 Home Screen (Feed)

- Top area:
 - App logo, search icon, notification icon.
 - Category filter chips horizontally scrollable.
- Stories row: user’s story + others’.
- Main feed: cards with post content, author (or “Anonymous”), college, category, timestamp, like/comment/share icons.

- Floating action button for post creation (if not in bottom nav).
- Pull-to-refresh, and “You’re all caught up” state.

6.5 Post Screen (Create/Edit)

- Text input area (multi-line).
- Media picker (gallery/camera; show selected thumbnails).
- Category dropdown/chips.
- Visibility: Followers, College, Circle-only.
- Anonymous toggle (if allowed by visibility).
- Post button and progress indication when uploading.

6.6 Post Detail Screen

- Full content with media, text, category, and meta-data.
- Like/comment/share.
- Comments list with input at the bottom.
- Report button and save/bookmark (optional).

6.7 Circle Screen

- Tabs for Anonymous, Events, Groups, Matching.

Anonymous tab:

- Similar to feed but only anonymous posts.
- “New anonymous post” button.

Events tab:

- Event list with upcoming at top.
- Filter by date and category.
- Event detail screen: full description, RSVP button, share, add to calendar (optional).

Groups tab:

- List of groups with join/request button.
- Group detail: description, members count, join status, open group feed and chat.

Matching tab:

- "Create card" CTA if user has no card.
- List/swipe of profile cards: show skills, looking-for, action buttons (Connect / Message).

6.8 Chat Screen

Chat List:

- List of recent chats with name, last message, time, unread badge.
- New chat button: search user or open group list.

Chat Detail:

- Messages in bubble style, time stamps.
- Input bar: text, emoji, attachment.
- Options: view profile, block, report, leave group.

6.9 Profile Screen

- Header: profile image, name, username, college, year/branch.
- Stats row: posts, followers, following.
- Edit profile button (for self).
- Tabs: Posts, Stories highlights (later), Circle activity (events/groups).
- Settings icon.

6.10 Settings Screen

- Account: email/phone, password change, logout.
- Privacy: profile visibility, block list.
- Notifications: toggles per category.
- Data & privacy: download my data, delete my account.
- Help & support: FAQs, report issue.

7. Design Style

Visual style:

- Modern, minimal, clean layout.
- Use dark and light theme with easy toggle in settings.
- Glassmorphism / subtle glass-like panels for cards and overlays.
- Smooth micro-animations: button presses, screen transitions, like animations.

Brand & feel:

- College-centric, youthful, yet professional.
- Avoid overly bright or aggressive colors that feel stressful.

- Typography simple and highly readable.

UX principles:

- Reduce clutter; one primary action per screen.
- Emphasize community and usefulness over endless entertainment.
- Consistent iconography and spacing across all screens.

8. Security and Privacy Rules

8.1 Post Visibility Rules

- Define clear visibility options:
 - Followers-only.
 - Entire college community.
 - Circle-only (college community within Circle features).
- Anonymous posts visible only inside the appropriate scope (college/Circle).
- No global public visibility (outside organization) in v1.

8.2 Messaging Rules

- Only users who follow each other or share a group can message by default (optional setting).
- Blocked users cannot send messages or see profile content.
- Rate limiting to prevent spam messages.
- Detect mass messaging / spam patterns and temporarily restrict.

8.3 User Reporting System

- Users can report: posts, comments, stories, chats, profiles, groups, events.
- Mandatory reason selection and optional description.
- Reports go into moderation system with ability to:
 - Hide content.
 - Warn or suspend users.
- Repeated violations lead to progressive penalties (temporary bans to permanent bans).

8.4 User Data Protection

- Store user passwords with strong hashing (e.g., bcrypt/argon2).
- Secure transmission via HTTPS everywhere.
- Minimize sensitive data stored (e.g., no plain-text DOB, no unnecessary PII).
- Regular backups of PostgreSQL database with secure access control.

8.5 Access Control

- Role-based access (RBAC):
 - User, Creator (optional), Group admin, College ambassador, Super admin.
- Backend routes guarded by auth middleware and RBAC checks.
- Separate admin APIs from user APIs with stricter authentication.

8.6 Data Usage Transparency

- Clear Privacy Policy explaining what data is collected, why, and how it's used.

- In-app explanation for sensitive features (anonymous posts, matching, hiring).
- Explicit consent for using data in skill-based hiring/monetization.

8.7 Data Deletion Policy

- Users can request account deletion from app.
- On deletion:
 - Anonymize personal data (name/username) from posts/comments but keep content if necessary, or hard-delete according to policy.
 - Remove tokens, sessions, and access rights.
- Retain logs and abuse reports for a limited period only for security/legal reasons.

8.8 Authentication Security

- Enforce strong password rules and optional 2FA later.
- Rate limiting on login/OTP endpoints.
- Refresh token with short-lived access token mechanism.
- Automatic logout of old sessions after password change.

8.9 Notification Privacy

- Push notifications should show minimal information (e.g., “You have a new message” instead of full message text, configurable).
- Option to disable preview content in notifications.
- Respect OS-level notification settings.

9. Tech Stack and Architecture

9.1 Frontend

- Framework: Flutter.
- Platforms: Android (initial), iOS later.
- State management: Provider / Riverpod / Bloc (to be finalized).
- Theming: Support for dark and light themes.

9.2 Backend

- Runtime: Node.js.
- Hosting: Railway (PaaS).
- Architecture: RESTful APIs for CRUD operations; WebSocket (e.g., Socket.io) for real-time chat and notifications.
- Authentication: JWT-based auth, with refresh tokens.
- Services:
 - Auth service.
 - User/profile service.
 - Feed/posts service.
 - Circle service (anonymous, events, groups, matching).
 - Chat service.
 - Notification service.

9.3 Database

- DB: PostgreSQL.

- Provider: Neon (serverless Postgres hosting).
- Schema (high-level entities):
 - Users, Profiles, Colleges/Organizations, Follows, Posts, Stories, Comments, Likes, Categories, Circles, Events, Groups, GroupMembers, Messages, Skills, MatchCards, Notifications, Reports, Roles.

9.4 Storage

- Provider: AWS S3.
- Use separate buckets/folders for: profile images, posts media, stories media.
- Store only URLs in DB, not raw files.

9.5 Caching Layer

- Cache: Redis.
- Use cases:
 - Session tokens or blacklist.
 - Frequently accessed feeds.
 - Rate-limiting counters.

9.6 Other Infrastructure

- Logging: centralized logs (e.g., Railway logs + external logging).
- Monitoring: basic uptime and performance monitoring.
- CI/CD: automatic builds and deployments for backend and Flutter app (later).

10. Non-Functional Requirements

Performance:

- Home feed should load first content within 1–2 seconds on good network.
- Chat messages should appear within near real-time.

Scalability:

- Design to support tens of thousands of users per college over time.
- Use pagination and lazy loading for all large lists.

Availability:

- Aim for >99% backend uptime.
- Graceful fallback when some services fail (e.g., feed loads but matching temporarily down).

Reliability:

- Idempotent APIs where possible.
- Safe error handling on frontend with clear messages and retry options.

Accessibility:

- Proper contrast in both dark and light themes.
- Support dynamic text sizes and basic screen reader compatibility.

11. Analytics and Success Metrics

Key metrics:

- Daily active users (DAU), weekly active users (WAU) per college.
- Average session duration (target: healthy engagement without extreme addiction).
- Number of posts per user per week (including anonymous).
- Events created and RSVPs per college.
- Matches initiated via Matching feature.
- Chat messages per user per week.
- Reports per 1000 posts (to monitor abuse).

12. Assumptions and Risks

Assumptions:

- Users have smartphones and stable internet.
- Colleges are willing to allow/encourage students to use such an app.
- Initial growth will be campus-by-campus via ambassadors.

Risks:

- Misuse of anonymous posting (bullying, harassment).
- Difficulty controlling addiction despite design intentions.
- Moderation load as communities grow.
- Legal and privacy compliance requirements in different regions.